Week 9: 26 Oct - 30 Oct 2026

Subject/الموضوع: Environmental science **Unit Title عنوان الوحدة: 4** **Grade الصف: 12 environmental**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Summative 1	I can examine environmental systems and decisions by applying concepts of sustainability, environmental ethics, matter and energy flow, ecosystem interactions, and biogeochemical cycles to real-world environmental scenarios.	8.2 Vary challenge level with differentiated tools, resources, and difficulty. 3.3 Guide processing with explicit steps, chunking, and interactive models. 6.2 Support planning with templates, strategy prompts, and modeled thinking.	From week 2 – week 6 on toddle and teams material
Lesson 2: **4.2 Community Diversity**	I can analyze community diversity by interpreting patterns of species abundance and distribution in a biological community.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 5.3 Build fluency with graduated scaffolds and models.	McGraw Hill Environmental Science — Chapter 4 Lesson materials available on **Teams — Term 1** Activities/submissions available through **Toddle**
Lesson 3: 4.2 Community Diversity	I can analyze how community complexity and productivity influence ecosystem resilience by using evidence from ecological models or data.	8.2 Vary challenge level with differentiated tools, resources, and difficulty. 8.1 Make goals clear, visible, and supported by short-term objectives. 5.3 Build fluency with graduated scaffolds and models.	McGraw Hill Environmental Science — Chapter 4 Lesson materials available on **Teams — Term 1** Activities/submissions available through **Toddle**
Lesson 4: 4.3 Communities Are Dynamic and Change Over Time	**I can analyze how communities change over time by comparing primary and secondary succession and predicting recovery after disturbance.**	8.2 Vary challenge level with differentiated tools, resources, and difficulty. 8.1 Make goals clear, visible, and supported by short-term objectives. 5.3 Build fluency with graduated scaffolds and models.	McGraw Hill Environmental Science — Chapter 4 Lesson materials available on **Teams — Term 1** Activities/submissions available through **Toddle**
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic *assessment information, topics, and expectations*; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams.		

	Effective preparation for graded formatives and summative requires students to actively engage with the learning materials provided in their Toddle class drive. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	Digital research tools, environmental diagrams, interactive models, and presentation resources are used to investigate environmental systems, trace energy and matter movement, analyze ecosystem relationships, and explain cause-and-effect patterns.
Cross-curricular links الروابط بين المواد الدراسية	**Biology: ecosystems, populations, food webs, producers, consumers, and living organisms.** **Chemistry: matter, atoms, molecules, pH, nutrients, and chemical energy.** **Mathematics: interpreting environmental models, patterns, quantities, and trophic relationships.** **Social Sciences: human use of natural resources and environmental decision-making.**
External Assessment Integration دمج التقييمات الخارجية	Students practice: • analyzing environmental models and scenarios; • identifying cause-and-effect relationships; • interpreting scientific evidence; • constructing systems diagrams; • predicting environmental consequences; • constructing evidence-based scientific explanations. • MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Critical Thinking: Analyze interconnected environmental systems and predict consequences.** **Research: Interpret scientific information and environmental evidence.** **Communication: Explain models and justify scientific reasoning.** **Collaboration: Construct models and discuss environmental relationships and solutions.**
Standards المعايير	HS-LS2-6 Evaluate claims, evidence, and reasoning that complex interactions in ecosystems maintain relatively consistent conditions but changing conditions may result in a new ecosystem.

	HS-LS2-1 Use quantitative and qualitative evidence to examine factors affecting populations and environmental systems. HS-ESS3-1 Connect natural-resource availability and environmental conditions to human activities and sustainable decision-making.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	**How can systems thinking help us understand how a change in one part of the environment affects the whole system?**
Lesson-specific success criteria	**Period 2 — 4.2 Diversity, Abundance & Community Structure** **Learning Target:** **I can analyze community diversity by interpreting patterns of species abundance and distribution in a biological community.** **MUST — DOK 1–2** **I can identify diversity, abundance, and patterns of distribution in a community.** **SHOULD — DOK 3** **I can analyze community data or diagrams to explain how species abundance and distribution patterns differ.** **COULD — DOK 4** **I can evaluate how a change in species abundance or distribution could affect the structure of a community.**
	Period 3 — 4.2 Resilience & Productivity **Learning Target:** **I can analyze how community complexity and productivity influence ecosystem resilience by using evidence from ecological models or data.** **MUST — DOK 1–2** **I can identify complexity, resilience, and primary productivity and the factors that affect productivity.** **SHOULD — DOK 3** **I can analyze evidence to explain how complexity and productivity influence community resilience.** **COULD — DOK 4** **I can predict and justify how a disturbance could affect the productivity and resilience of an ecosystem.**
	Period 4 — 4.3 Community Change & Succession **Learning Target:** **I can analyze how communities change over time by comparing primary and secondary succession and predicting recovery after disturbance.** **MUST — DOK 1–2** **I can identify primary succession, secondary succession, disturbance, and climax community.** **SHOULD — DOK 3** **I can compare primary and secondary succession and explain how a community changes after disturbance.** **COULD — DOK 4** **I can predict and justify how the type or severity of a disturbance could affect the speed and pattern of community recovery.**

	Lesson 4 MUST — DOK 1–2 I can identify a UAE sustainability challenge and the ecological factors involved. SHOULD — DOK 3 I can analyze how biodiversity, species interactions, and community resilience are connected to the UAE sustainability challenge. COULD — DOK 4 I can evaluate and justify a realistic sustainability solution for the UAE using ecological evidence.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Global Understanding التفاهم العالمي Heritage التراث **Belonging** • **Students recognize their role in protecting and responsibly managing UAE ecosystems and natural resources.** **Global Understanding** • **Students connect UAE ecosystems to wider global patterns of energy flow, matter cycling, biodiversity, and environmental interdependence.** **Heritage** • **Students connect environmental systems and resource use to UAE landscapes, agriculture, water use, coastal environments, and traditional relationships with the natural environment.**